IN 22 EN3330 Assignment 03

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1. Consider the constrained problem

$$\min_x x^2 \quad \text{subject to } x \geq 1. \tag{1}$$

Writing the constraint as $h(x) = 1 - x \leq 0$, the Lagrangian is

$$L(\mu, x) = x^2 + \mu(1 - x), \qquad \mu \geq 0. \tag{2}$$

(a) Plot $L(\mu, x)$ as a function of x for $\mu = 0, 1, 2, 3$ on the same axes. Mark the minimizer

$$x(\mu) = \arg\min_x L(\mu, x) \tag{3}$$

of each curve and describe how it moves as μ increases. [marks 30]

(b) Derive the dual function

$$d(\mu) = \min_x L(\mu, x) \tag{4}$$

analytically and plot it against μ. Identify

$$\mu^* = \arg\max_{\mu \geq 0} d(\mu) \tag{5}$$

and the dual optimal value d^*. [marks 30]

(c) Plot the primal function

$$g(x) = \max_{\mu \geq 0} L(\mu, x) \tag{6}$$

for $x \in [-1, 3]$. Show graphically that $g(x) = x^2$ for feasible x and $g(x) = +\infty$ for infeasible x. Hence explain why

$$\min_x g(x) \tag{7}$$

is equivalent to problem (1). [marks 40]

2. Consider the problem and Lagrangian in question 1. Let $x^* = 1$ and $\mu^* = 2$ be the optimal primal and dual solutions found there.

(a) Show that (x^*, μ^*) is a saddle point of L, i.e.

$$L(\mu, x^*) \leq L(\mu^*, x^*) \leq L(\mu^*, x) \quad \forall\, \mu \geq 0,\ \forall\, x. \tag{8}$$

Follow these steps:

(i) Compute $L(\mu^*, x^*)$.

(ii) Compute $L(\mu, x^*)$ and show that it does not depend on μ. Hence verify the left inequality.

(iii) Show that $L(\mu^*, x) = (x - 1)^2 + 1$. Hence verify the right inequality.

(iv) Plot $L(\mu, x^*)$ against μ and $L(\mu^*, x)$ against x (two small graphs) and mark the common value $L(\mu^*, x^*)$.

[marks 50]

(b) The optimality argument for the right inequality can be written as

$$f(x^*) \overset{(1)}{=} L(\mu^*, x^*) \overset{(2)}{\leq} L(\mu^*, x) \overset{(3)}{=} f(x) + \mu^* h(x) \overset{(4)}{\leq} f(x), \tag{9}$$

for any feasible x. Give a one-line reason for each step (1)–(4). What does the left inequality tell you about $h(x^*)$ if $L(\mu, x^*)$ must stay finite as $\mu \to \infty$?

[marks 25]

(c) Verify that the complementary slackness condition

$$\mu^* h(x^*) = 0 \tag{10}$$

holds. Now change the constraint to $x \geq -1$. Is the unconstrained minimizer $x = 0$ of x^2 feasible? Is the constraint active there? What value would you expect for μ^*, and why?

[marks 25]

3. **Weak duality.** Consider the general problem

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) \quad \text{subject to } h(\boldsymbol{x}) \leq 0, \tag{11}$$

with Lagrangian $L(\mu, \boldsymbol{x}) = f(\boldsymbol{x}) + \mu h(\boldsymbol{x})$ and dual function

$$d(\mu) = \min_{\boldsymbol{x}} L(\mu, \boldsymbol{x}), \qquad \mu \geq 0. \tag{12}$$

Let p^* denote the optimal value of problem (11).

(a) Let $\mu \geq 0$ and let $\boldsymbol{x}$ be any feasible point (so $h(\boldsymbol{x}) \leq 0$). Give the reason for each of the two inequalities in

$$d(\mu) \overset{(1)}{\leq} L(\mu, \boldsymbol{x}) = f(\boldsymbol{x}) + \mu h(\boldsymbol{x}) \overset{(2)}{\leq} f(\boldsymbol{x}). \tag{13}$$

Conclude that $d(\mu) \leq p^*$ for every $\mu \geq 0$. Then check this numerically for question 1: using $d(\mu)$ from Q1(b), verify that $d(\mu) \leq p^* = 1$ for $\mu = 0, 1, 2, 3$. [marks 30]

(b) For question 1, sketch $L(\mu, x) = x^2 + \mu(1 - x)$ as a function of μ for $x = 0, 1, 2$ on the same axes (these are straight lines). Draw the lower envelope of the three lines, which approximates $d(\mu)$. Is it concave or convex? Use this picture to explain in one or two sentences why $d(\mu)$ is concave in μ for *any* f, even a non-convex one. [marks 20]

Submission

- Upload a report as a pdf file named as "Yourindexno_EN3330_A03.pdf". Include the index number and the name within the report as well.
- The interpretation of results and the discussion are important in the report.
- Pay careful attention to formatting such as font size, spacing, and margins.
- Include a title page with necessary information (e.g., title, author, date, index no).
- Use consistent and professional formatting throughout the document.
- Plagiarism will be checked and in cases of plagiarism, an extra penalty of 50% will be applied. In case of copying from each other, both parties involved will receive a grade of zero for the assignment. Academic integrity is of utmost importance, and any form of plagiarism[1] or cheating will not be tolerated.
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[1] https://en.wikipedia.org/wiki/Plagiarism